POKHARA UNIVERSITY

Level: Bachelor	Semester: Spring	Year	: 2024
Programme: BE		Full Marks	: 100
Course: Engineering Economics (New)		Pass Marks	: 45
		Time	: 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) List out the principles of Economy. **[5]**
1. b) How you understand supply and demand. **[5]**
1. c) Describe about manufacturing cost and Non-manufacturing cost. **[5]**

2. a) A man is planning to retire in 25 years. He wishes to deposit regular money every 3 months until he retires so that he will receive annual payments of Rs. 4,50,000 after the first year of his retirement for the next 10 years. How much must he deposit if the interest rate is 8%, compounded monthly? **[8]**
2. b) Explain about simple interest, compound interest, nominal interest and effective interest in your own word. **[7]**

3. a) Calculate the simple and discounted payback period for the project when MARR is 20%. **[8]**
- Initial investment: Rs 25000
- Annual Revenue: Rs 8000
- Salvage Value: Rs 5000
- Expected Life: 5 year.

3. b) Compute IRR and suggest the investment decision for the project given below: **[7]**
- Initial investment: Rs 90000
- Annual Revenue: Rs 27000
- Annual Expenses: Rs 8000
- Salvage Value: Rs 10000
- Expected Life: 6 year
- MARR: 15%

4. a) Using co-terminated assumption recommend the best project taking study period as 5 years. **[8]**

Project	**A**	**B**
Initial Investment (Rs)	3,50,000	5,00,000
Annual Revenues (Rs)	1,30,000	1,75,000
Annual cost (Rs)	15,000	25,000
Salvage value (Rs)	35,000	50,000
Useful life	5 years	8 years
MARR	**10%**	

OR

Evaluate the following two feasible investment A and B having different useful lives, if MARR is 10% per year. Use PW method with repeatability assumption.

Project: A	**Project: B**
Initial investment: Rs 50000	Initial investment: Rs 150000
Annual Revenue: Rs 25000	Annual Revenue: Rs 70000
Annual Expenses: Rs 3000	Annual Expenses: Rs 2000
Salvage Value: Rs 15000	Salvage Value: Rs 40000
Expected Life: 3 year	Expected Life: 5 year

4. b) Suppose a company has decided to purchase a new machine having initial cost of Rs 18,000 by expecting 8 years useful life & have operating cost of Rs 3000 in 1st year. For remaining years, operating cost increase each year by 15% over previous year's operating cost & salvage value declines each year by 20% from the previous year's salvage value. Find the economic service life of this new machine when MARR is 12% per year. **[7]**

5. a) Perform Sensitivity analysis using PW method over a range of (+ or -) 20% in initial investment, net annual revenue and salvage value. **[8]**

- Initial investment: Rs 200000
- Annual Revenue: Rs 50000
- Annual Expenses: Rs 5000
- Salvage Value: Rs 25000
- Expected Life: 10 year
- MARR: 12% per year

5. b) What do you understand about break-even analysis? State the procedures to identify break even point. **[7]**

OR

If you purchase Jagadamba cement at the rate Rs 650 and selling at the rate of Rs 900, you are paying shop rent change Rs 10000 and Rs 12000 for the helper salary. What is the break-even quantity that you must sold. Let us say you have 10000 bags of cement is in stock.

6. a) A grinding machine has initial cost & annual saving per year of Rs. 40,000 & Rs. 20,000 respectively with annual O & M cost of Rs. 6,000. It will depreciate by MACRS method & will have salvage value of Rs. 4,000 at the end of 5th year. Estimate BTCF & ATCF. The company pays income tax @ 40%. Evaluate the profitability by FW method if interest rate is 12%. Also, calculate the interest rate by IRR analysis. **[8]**

6. b) Discuss about different methods of financing including its merits and demerits. **[7]**

7. Write short notes on: (Any two) [2x5]

a) Inflation

b) Risk Analysis

c) Independent and contingent project

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Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What do you understand about independent and dependent projects? (5)
1. b) Define inflation. What are its causes? (5)
1. c) How do you understand the economic service life of a machine? (5)

2. a) Define time value of money. Suppose that you make the monthly deposits of Rs 5,000 each into a bank account that pays an interest rate of 8% compounded weekly for 5 years. After 5 years, interest rate changes to 6% per year, how much money will you have accumulated in this bank account at the end of 8 years? (8)
2. b) What is Supply & demand? Explain about principles of engineering economics. (7)

3. a) Compute IRR and suggest the investment decision for the project given below: (8)

- Initial investment: Rs 90,000
- Annual Revenue: Rs 27,000
- Annual Expenses: Rs 8,000
- Salvage Value: Rs 10,000
- Expected Life: 6 year
- MARR: 15%

OR

Find the simple and discounted payback periods and justify investment with the given cash flow information:

- Initial investment: Rs 400,000
- Annual benefit: Rs 150,000
- Annual Cost: Rs 30,000
- Salvage value: Rs 100,000
- Useful life: 5 years
- Interest rate: 10% per year.

3. b) Explain about manufacturing cost and non-manufacturing cost. (7)

Github.com/theharshchaudhary

4. a) Using co-terminated assumption recommend the best project taking study period as 5 years. **(8)**

Project	**A**	**B**
Initial Investment (Rs)	3,50,000	5,00,000
Annual Revenues (Rs)	1,30,000	1,75,000
Annual cost (Rs)	15,000	25,000
Salvage value (Rs)	35,000	50,000
Useful life	5 years	8 years
MARR	**10%**	

OR

Evaluate the following two feasible investment A and B having different useful lives, if MARR is 10% per year. Use PW method with repeatability assumption.

Project: A	**Project: B**
Initial investment: Rs 50000	Initial investment: Rs 150000
Annual Revenue: Rs 25000	Annual Revenue: Rs 70000
Annual Expenses: Rs 3000	Annual Expenses: Rs 2000
Salvage Value: Rs 15000	Salvage Value: Rs 40000
Expected Life: 3 year	Expected Life: 5 year

4. b) Explain different project funding mechanism in Nepal for public projects. [7]

5. a) Perform Sensitivity analysis using PW method over a range of (+ or -) 20% in initial investment, net annual revenue and salvage value. **(8)**

- Initial investment: Rs. 200000
- Annual Revenue: Rs. 50000
- Annual Expenses: Rs. 5000
- Salvage Value: Rs. 25000
- Expected Life: 10 year
- MARR: 12% per year

5. b) Suppose a company has decided to purchase a new machine having initial cost of Rs 18,000 by expecting 8 years useful life & have operating cost of Rs 3000 in 1st year. For remaining years, operating cost increase each year by 15% over previous year's operating cost & salvage value declines each year by 20% from the previous year's salvage value. Find the economic service life of this new machine when MARR is 12% per year. [7]

6. a) Define depreciation. What are the causes of it? If a machine costing Rs. 1,50,000 is purchased by expecting salvage value of Rs. 40,000 at the end of 6th year. Calculate depreciation amount for each year by: **(8)**

i. SOYD
ii. Declining balance.

6. b) Evaluate whether the project is feasible for investment or not using different types of BCR? **(7)**

- Initial investment: Rs. 400000
- Annual benefit: Rs. 75000
- Annual Disbursement: Rs. 25000
- Salvage value: Rs. 100000
- Useful life: 10 years
- Interest rate: 10% per year.

Write short notes on: **(Any two) (2x5)**

a) Depreciation by Sinking Fund method
b) Monte Carlo Simulation
c) Simple, Compound and Nominal Interest

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Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1.a) Discuss Engineering Economics and describe Principles and application 7 of Engineering Economics. [7]

1.b) Define with a suitable example about the opportunity cost, marginal cost 8 and sunk cost? Also explain manufacturing and non-manufacturing cost. [8]

2.a) Ghanshyam is planning for his retirement. He is going to place a deposit Rs. 30000 per month in Citizen Investment Trust (CIT). CIT will give an interest rate of 7.5% compounded quarterly. If he continues his deposit for 15 years from now how much money he will have in his account? [7]

OR

Mr. Gurung needs Rs. 1500000 at the end of 15th year for the study of his daughter. How much money does he need to deposit every year in a bank account for 10 years if interest rate is 9% compounded annually?

2.b) Calculate both types of BCR using AW and FW formulation for the following Project. [8]

Initial Investment	500000
Annual Revenue	10000
Annual Cost	15000
Useful Life	8 years
Salvage Value	50000
MARR	12%

3.a) Compute IRR and suggest the investment decision and also draw unrecovered investment diagram for the project whose cash-flow given below: [8]

Initial investment: Rs 90000
Annual Revenue: Rs 27000
Annual Expenses: Rs 8000
Salvage Value: Rs 10000
Expected Life: 6 years
MARR: 15%

Github.com/theharshchaudhary

3.b) Evaluate whether the project is feasible for investment or not using Payback period method? Find both types of Payback period. [7]

Initial investment: Rs. 400000
Annual benefit: Rs. 75000
Annual cost: Rs. 25000
Salvage value: Rs. 100000
Useful life: 10 years
Interest rate: 10% per year

4.a) Perform Sensitivity analysis of the following project over a range of (+-)30% in a) Initial Investment b) Net annual income and c) Useful life using PW formulation. MARR=10%. Draw Sensitivity diagram also. [8]

Initial Investment	20000
Annual Revenue	5000
Annual Cost	1000
Useful life	8 years
Salvage Value	2000

4.b) From the following information, find economic service life of an assets. [8]
Initial Cost: Rs. 50,000

Annual Operating cost: Rs 10,000 for the first year and increases by 15% each year
Salvage value: Declining each year by 20% from the previous year's salvage value
Useful life: 7 yrs
MARR: 15%

OR

Explain the reasons for replacement of assets. The Annual Equivalent Cost (AEC) of the defender and challenger are given in the table below. What is the best replacement strategy? Use MARR: 12% per year and the planning horizon of the project is 8 years.

End of Year (n)	1	2	3	4	5	6
$(AEC)_D$ (Rs)	5,300	5,250	5,400	5,750	6,200	6,550
$(AEC)_C$ (Rs)	7,700	6,150	5,700	5,600	5,675	5,800

5.a) Compute the annual depreciation allowance and resulting book value. [8]
using

i. SOYD method
ii. Sinking fund method if the Initial cost of equipment is Rs 10,000, Salvage value is Rs 2,000 and useful life is 5 years. Assume MARR -12%.

5.b) What do you understand by capital budgeting? Explain the different methods of financing an engineering project with its advantages and disadvantages.

6.a) What are the reasons of inflation? Explain actual dollar and constant 7 dollar. [7]

6.b) Evaluate the following two feasible investment A and B having different useful lives, if MARR is 10% per year. Use PW method with repeatability assumption. [8]

Project: A	Project: B
Initial Investment: Rs. 50000	Initial Investment: Rs. 150000
Annual Revenue: Rs. 25000	Annual Revenue: Rs. 70,000
Annual Expenses: Rs. 3000	Annual Expenses: Rs. 2000
Salvage Values: Rs. 15000	Salvage Value: Rs. 40,000
Expected Life: 3 years	Expected Life: 5 year

7. Write short notes on: (Any two) [2x5]

a. Causes of project risk
b. Value added tax
c. Time value of money